SoCSE AI Day 2026
Prototype Evaluation

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Low-Light Image Enhancement: Hybrid Retinex Pipeline
Problem Statement 1 | NexGen Creators

Abhishek D Nagoor - 1RUA24CSE0009
Abubakar Khan – 1RUA24CSE0010
Aditi M – 1RUA24CSE0019

Problem Statement

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The problem

- Photos taken in low light are dark, noisy, colour-shifted and unevenly lit, so detail is lost.
- Fixed enhancement constants (one gamma, one denoise strength) cannot suit every scene.
- RetinexNet alone gives an orange cast on night scenes, speckle noise on dim streets and clips 6.6% of pixels.
- The supplied code was hard-wired to .cuda() and its training shuffle was a no-op.

Problem in one line

Build a robust, adaptive low-light enhancement prototype (Hackathon Problem Statement 1) that combines RetinexNet with classical computer vision, runs on a free CPU host, and is judged with full-reference and no-reference metrics.

Relevance and context

Where: Hackathon Problem Statement 1, evaluated on LOL low-light images (3 real test images used so far; eval15 for submission).

Why Retinex: an image is reflectance times illumination; correcting illumination while keeping reflectance restores detail.

Constraints: CPU-only hosting, a public demo URL and a latency target of about 10 s per image.

Noise
speckle in dim regions

Colour cast
orange tint on night scenes

Uneven light
deep shadows and bright spots

Clipping
overexposed highlights

Objectives

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Primary goals

- Replace fixed constants with adaptive, per-image parameters (advanced requirement #1).
- Combine a classical Retinex pipeline with pretrained RetinexNet for robustness.
- Make RetinexNet CPU/GPU-agnostic without changing the architecture or weights.
- Weight outputs by a no-reference quality score where ground truth is absent.
- Document ablations, failures and limitations honestly.

Key functionalities and deliverables

- **Gradio app:** before/after slider, modes (hybrid, classical, deep learning, compare all), 14 intermediate views, metrics, per-stage latency.
- **evaluate.py:** method comparison, ablation, eval_P1_<TEAM>.md report.
- **train.py:** fixed fine-tuning with real shuffle and best-checkpoint saving.
- **Docs + packaging:** README, design, failure analysis, architecture diagram, Dockerfile.

How success is measured

Full-reference (with ground truth)

- PSNR (dB) and SSIM
- Delta E 2000 colour error
- Hue shift

No-reference (real images)

- Noise ratio (lower is better)
- Edge preservation (higher is better)
- Newly overexposed pixels (%)
- No-reference quality score

Operational targets

- About 10 s per image on CPU
- Zero newly clipped highlights
- Graceful errors for empty, tiny, black, grayscale inputs

Methodology & Workflow

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1. **Code review**
   Read RetinexNet model.py and training/test code; found .cuda() hard-wiring and a no-op shuffle.
2. **Environment check**
   Verified torch, OpenCV and weights; fetched the 9200-step Decom and Relight checkpoints from upstream.
3. **CPU/GPU-agnostic RetinexNet**
   Same architecture, so pretrained weights load unchanged and run on CPU.
4. **Classical pipeline**
   Guided-filter illumination, Retinex decomposition, noise-aware denoising, CLAHE, colour and highlight correction.
5. **Scene analysis + adaptive parameters**
   Darkness, noise, non-uniformity and colour cast map to per-image parameters.
6. **Hybrid fusion**
   Log-domain illumination fusion gives three candidates, merged per pixel by quality weights.
7. **Evaluate, ablate, package**
   12 synthetic pairs and 3 real images; remove one component at a time; Gradio app and docs.

Tools and libraries

- **Python + PyTorch:** CPU wheel from PyPI (CPU index unreachable)
- **OpenCV + NumPy:** classical CV; NumPy 2 dtype fix
- **RetinexNet:** Decom/9200.tar and Relight/9200.tar
- **Gradio >=6,<7:** web app (theme/css moved to launch())
- **Docker / HF Spaces:** deployment route
- **Planned:** lpips and pyiqa for official LPIPS and NIQE

Methodology & Workflow

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Input image
dark / noisy photo

Scene analysis
darkness, noise, non-uniformity, colour cast

Adaptive parameter selector
choose_params: dark / dim / normal rules

Candidate 1: Classical
Guided-filter Retinex, adaptive gamma, noise-aware denoise, CLAHE, white balance, saturation, highlight roll-off

Candidate 2: RetinexNet
Pretrained DecomNet + RelightNet (9200-step checkpoints) on CPU; MSRCR kept as baseline in classical module

Candidate 3: Fused illumination
Classical and learned illumination estimates merged in the log domain

Quality-based per-pixel merge
no-reference quality scorer weights the three candidates pixel by pixel

Output: enhanced image + metrics table + 14 intermediate visualisations

Latency (640 px, CPU): classical ~0.13 s, RetinexNet ~3.3 s, hybrid ~3.8 s

Ablation toggles: each component can be switched off to measure its effect

Input handling: empty, tiny, black and grayscale inputs return clear errors

Prototype Demonstration

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Cloud deployment / demo link

- **Public URL:** [add after deploying to Hugging Face Spaces]
- **Status:** not live yet; Dockerfile and requirements.txt are packaged.
- **Local check:** fresh start returned a 640 px hybrid result in 8.4 s on one CPU core, no errors.

Plan for live demo

1. Upload a dark photo (night scene or dim street)
2. Drag the before/after slider in hybrid mode
3. Compare all three: classical, RetinexNet, hybrid
4. Open intermediate views and per-stage latency
5. Read the metrics table; add ground truth for PSNR/SSIM
6. Try empty, tiny, black or grayscale input

What the app offers

- Before/after slider for every mode
- Mode selector: hybrid, classical, deep learning, or compare all three
- 14 intermediate visualisations of the pipeline
- Metrics table and per-stage latency breakdown
- Optional ground-truth upload for PSNR and SSIM
- Error handling for empty, tiny, black and grayscale inputs

What the audience should see

- RetinexNet alone: orange cast on the night scene, speckle on the street scene.
- Hybrid avoids both, with 0% newly clipped pixels.
- Cost: about 4 to 8 s on CPU versus 0.13 s for classical.

Results & Ablation

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Synthetic pairs (12, own darkened photos)

Method	PSNR (dB)	SSIM
Classical	**17.14**	0.452
RetinexNet alone	15.04	0.410
Hybrid	16.46	**0.530**

Real LOL images (3 unseen, no ground truth)

Metric	Classical	RetinexNet	Hybrid
Noise ratio (lower)	0.021	0.030	**0.018**
Edge preservation	0.896	0.865	**0.921**
New overexposure	**0%**	6.56%	**0%**
Latency (640 px)	~0.13 s	~3.3 s	~3.8 s

Reading it: hybrid trails classical by 0.68 dB PSNR but leads SSIM by 0.078 and has the lowest noise. RetinexNet alone clips 6.6% of pixels and shows an orange cast.

Helps

- Denoising: biggest contributor
- Adaptive parameters replace fixed constants (req. #1)
- Early test: +0.7 to 1.6 dB PSNR; noise ratio ~0.17 to 0.11-0.14

Hurts here

- CLAHE costs a little SSIM/PSNR
- Colour-correction step costs a little too
- Seen on very noisy synthetic data; may differ on cleaner images

Neutral

- Highlight guard: no effect on the synthetic set
- Real images: hybrid shows 0% newly clipped pixels
- Exposure deliberately not tuned, to avoid overfitting

Mixed

- Quality-based fusion: SSIM 0.47 to 0.53
- But PSNR goes down
- Trade-off: structure versus pixel accuracy

Conclusion & Future Scope

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Key achievements

- Working hybrid prototype, tested locally and packaged as a zip.
- Hybrid: lowest noise (0.018) and best edge preservation (0.921) on real images, 0% new clipping.
- Synthetic SSIM 0.530 vs 0.452 (classical) and 0.410 (RetinexNet).
- RetinexNet made CPU-capable; 8.4 s fresh-start run on one core.
- Ablation and four documented failure cases with causes.

Current limitations

- Hybrid PSNR trails classical (16.46 vs 17.14); synthetic data is a smoke test, not LOL.
- Hazy in deep shadows and somewhat desaturated on very dark scenes.
- Official NIQE and LPIPS not computed; no fine-tuned weights.
- Thresholds (white balance, dark/dim/normal) are heuristic, not cross-validated.
- About 4 to 8 s per image on CPU; the 10 s target is tight.

Future scope and next steps

Close the evaluation gap

- Run evaluate.py --data_dir on LOL eval15
- Install lpips and pyiqa for official LPIPS and NIQE

Improve the model

- Fine-tune with train.py on LOL training data
- Cross-validate thresholds; reduce shadow haze and desaturation

Deploy and extend

- Hugging Face Space from the Dockerfile for a public URL
- Explore learned fusion weights and a colour metric that does not penalise cast removal

References

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Papers and methods

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- K. He, J. Sun, X. Tang, "Guided Image Filtering," IEEE TPAMI, 2013.
- D. Jobson, Z. Rahman, G. Woodell, "A Multiscale Retinex for Bridging the Gap Between Color Images and the Human Observation of Scenes," IEEE TIP, 1997 (MSRCR).
- K. Zuiderveld, "Contrast Limited Adaptive Histogram Equalization," Graphics Gems IV, 1994 (CLAHE).
- A. Mittal et al., "Making a Completely Blind Image Quality Analyzer," IEEE Signal Processing Letters, 2013 (NIQE).
- R. Zhang et al., "The Unreasonable Effectiveness of Deep Features as a Perceptual Metric," CVPR, 2018 (LPIPS).

Dataset and code

- Dataset: LOL low-light dataset [add URL]; eval15 split for submission.
- RetinexNet PyTorch master code and pretrained checkpoints: upstream repository [add URL].
- Synthetic pairs: 12 photos darkened by the team (not a benchmark).

Tools and libraries documentation

- PyTorch, OpenCV, NumPy: official documentation [add URLs].
- Gradio 6 documentation (theme and css in launch()).
- pyiqa and lpips packages for NIQE and LPIPS (planned).
- Please verify each citation before submission.